

A.U.R.A. SENTINEL

Autonomous Unified Risk Analytics Platform

AI-Powered Continuous Cyber Risk Quantification & Investment Optimization Platform

PROBLEM STATEMENT	SUBMITTING TEAM	THEME & CATEGORY
SIH26105 Enterprise Defense	ByteForce_1 Team ID: 180219	Blockchain & Cyber Software Track

DOCUMENT: Technology, Research & Implementation Resource Pack

This resource pack provides SIH evaluators with verified algorithmic proofs, architectural justification, source code file links, and official academic citations for all implemented modules.

1. Technology Implementation Matrix

The matrix below maps all implemented technologies to their functional purpose, codebase role, and official external documentation:

Technology	Purpose in AURA	How AURA Uses It	Status	Official Reference
YOLOv8 (Ultralytics)	Object Detection	Used for computer-vision based operator presence detection & boundary surveillance.	IMPLEMENTED	docs.ultralytics.com
MITRE ATT&CK®	Attack Technique Mapping	Maps open listening ports and anomalous processes to standardized techniques (T1046, T1021, T1059).	IMPLEMENTED	attack.mitre.org
psutil + OS APIs	System & Network Telemetry	Streams kernel CPU, memory, disk I/O, process lists, and network listening sockets on 1s intervals.	IMPLEMENTED	psutil.readthedocs.io
0/1 Knapsack DP	Security Investment Optimization	Selects optimal non-fractional defense controls under budget constraint via O(N-W) dynamic programming.	IMPLEMENTED	cs.odu.edu/~zeil
SHA-256 + Merkle (FIPS 180-4)	Tamper-Evident Audit Ledger	Hashes every telemetry alert and decision with SHA-256 and Merkle roots for immutable forensic proof.	IMPLEMENTED	csrc.nist.gov
Groq SDK / LLM	AURA Voice / AI Assistant	Provides sub-second conversational risk guidance and board-level risk explanations via Groq LPU inference.	IMPLEMENTED	console.groq.com

2. AURA Core Pipeline Architecture

The 9 operational stages converting raw machine telemetry into auditable board-level cybersecurity investments:

- **1. SYSTEM TELEMETRY** [psutil + OS Sockets]: Asynchronous host ingestion of CPU, RAM, disk I/O, open ports, and processes.
- **2. COLLECT & NORMALIZE** [FastAPI + Pydantic]: Multi-source log ingestion, JSON schema normalization, and timestamp synchronization.
- **3. THREAT ANALYSIS** [MITRE ATT&CK;]: Maps active listening sockets to MITRE ATT&CK; technique IDs (T1046, T1021, T1059).
- **4. RISK SCORING** [Statistical Scorer]: Multi-factor composite scoring calculating technical cyber severity on a 0–100 scale.
- **5. FINANCIAL EXPOSURE** [Actuarial Loss Model]: Translates technical severity into modeled rupee liability (■8L–■15L illustrative output).
- **6. WHAT-IF WARGAME** [Sensitivity Engine]: Interactive scenario simulation modeling shifts in open ports, threat signals, and load.
- **7. 0/1 KNAPSACK DP** [Dynamic Programming]: Discrete non-fractional budget optimizer maximizing risk mitigation strictly within budget (W).
- **8. AI EXPLANATION** [Groq LPU (Llama-3)]: Sub-second contextual risk justification and conversational operator Q&A; assistance.
- **9. BLOCKCHAIN AUDIT** [SHA-256 + Merkle]: Tamper-evident cryptographic ledger sealing all decisions with proof-of-work validation.

3. Implemented AURA Modules (12 Modules)

Summary of all 12 operational technical modules present in the current build:

Module	Technology	Functional Capability	Source File Location
01. System Monitoring	psutil + OS APIs	Real-time CPU, RAM, disk, and process telemetry.	Backend/main.py
02. Risk Intelligence	Statistical Scorer	Composite cyber risk scoring on 0–100 scale.	Ai_Engine/risk_scorer.py
03. Attack Surface	MITRE ATT&CK;	Port scanner & MITRE adversary technique mapping.	Ai_Engine/threat_detector.py
04. Financial Risk	Loss Modeling	Estimated financial exposure (■8L–■15L prototype).	Ai_Engine/financial_engine.py
05. What-If Engine	Wargame Modeler	Dynamic threat vector & compute stress simulation.	Backend/main.py
06. Investment Optimizer	0/1 Knapsack DP	Discrete budget-optimal cyber defense allocation.	Backend/investment_optimizer.py
07. Blockchain Ledger	SHA-256 + Merkle	Tamper-evident audit chain with tamper detection.	Backend/blockchain_ledger.py
08. AURA Voice / AI	Groq Cloud SDK	Conversational risk explanation & natural query.	Ai_Engine/groq_provider.py
09. YOLOv8 Detection	Ultralytics YOLO	Workstation presence surveillance & object tracking.	Backend/yolov8n.pt
10. MediaPipe Gestures	MediaPipe Hands	Touchless emergency lock & navigation gestures.	Ai_Engine/gesture_controller.py
11. Sensitive File Scan	Heuristic Parser	Detects leaked API keys, tokens, & private keys.	Backend/main.py

12. Quarantine Vault	send2trash / Vault	Safely isolates and restores suspicious host files.	Backend/main.py
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4. Code / Source Traceability

Direct mapping of each major feature to actual verified source files in the project:

- **0/1 Knapsack Budget Optimizer**

Impl: Backend/investment_optimizer.py | **Verif:** Backend/test_knapsack.py

Description: 2D DP recurrence table, backtracking, and discrete constraints. 10 passing unit tests (0.006s).

- **Blockchain Audit Ledger**

Impl: Backend/blockchain_ledger.py | **Verif:** Database/blockchain_ledger.json

Description: SHA-256 hashing, parent-hash verification, Merkle root creation, and tamper detection.

- **Financial Risk Engine**

Impl: Ai_Engine/Risk_engine/financial_engine.py | **Verif:** Backend/services/financial_engine.py

Description: Translates CVSS severity into model-based financial loss liability ranges.

- **Threat & Port Intelligence**

Impl: Ai_Engine/Detection/threat_detector.py | **Verif:** Backend/main.py

Description: Network socket scanner mapping open ports to MITRE ATT&CK; techniques (T1046, T1021).

- **System Telemetry Engine**

Impl: Backend/main.py | **Verif:** Ai_Engine/Anomaly/system_anomaly.py

Description: Kernel telemetry stream via psutil (CPU, RAM, disk, process tables).

- **YOLOv8 & MediaPipe Vision**

Impl: Backend/main.py | **Verif:** Ai_Engine/Detection/gesture_controller.py

Description: Real-time presence lock and touchless gesture control.

- **Sensitive File & Quarantine**

Impl: Backend/main.py (/api/file-security/*) | **Verif:** Frontend/src/components/modules/FileSecurity.jsx

Description: Heuristic token scanner and safe file quarantine with restore.

- **Frontend & Dashboard Mockup**

Impl: Frontend/src/App.jsx | **Verif:** Frontend/src/components/showcase/

Description: React 19 client with Judge Mode, Executive SOC Console, and live module views.

5. Current Implementation vs. Future Roadmap

CURRENTLY IMPLEMENTED (Active in Build):

All 12 technical modules listed above, 0/1 Knapsack DP algorithm, SHA-256 Merkle blockchain ledger, What-If simulator, and FastAPI/React architecture are fully operational in code.

FUTURE ROADMAP (Planned Enterprise Evolutions):

- **FAIR Quantitative Loss Methodology:** Factor Analysis of Information Risk (Open FAIR™) Monte Carlo probability modeling.
- **NIST CSF 2.0 Mapping:** Automated alignment across Govern, Identify, Protect, Detect, Respond, Recover tiers.
- **NIST SP 800-30 Assessment:** Federal threat event cataloging and structured likelihood/impact matrices.
- **CIS Controls v8 Alignment:** Automated compliance checks against CIS benchmark endpoint configurations.
- **ISO/IEC 27001 Control Mapping:** Annex A technical control cross-referencing and compliance audit logging.
- **Adaptive Bayesian AI Forecasting:** Long-term telemetry sequence modeling to predict vulnerability emergence.

6. Claim Safety & Evaluator Disclosures

- **No Certification Claims:** AURA Sentinel does not claim accredited compliance certifications (e.g. 'ISO certified', 'NIST compliant', 'DPDP certified') as these require independent accredited auditing.
- **Illustrative Prototype Outputs:** Financial figures (■8L–■15L) and risk scores (82/100) are explicitly labeled 'Illustrative Prototype Output'. They represent model-based simulations rather than empirical insurance claims.
- **0/1 Knapsack Optimality:** The algorithm produces optimal portfolios strictly 'among configured candidate investments under specified constraints'. It does not claim enterprise-wide guaranteed global optimality.
- **Tamper-Evident Ledger:** Labeled as a 'Tamper-Evident Audit Ledger' using SHA-256 and Merkle trees on a local node; it does not claim to be 'tamper-proof' or decentralized.

7. Evaluator Quick View (1-Page Summary)

AURA SENTINEL AT A GLANCE — SIH 2026 EVALUATOR CHEAT-SHEET

WHAT AURA CURRENTLY DOES (100% IMPLEMENTED):

1. Ingests kernel-level system telemetry (CPU, RAM, disk, processes, open ports) via psutil.
2. Maps listening ports and anomalous activities to standardized MITRE ATT&CK; techniques (T1046, T1021).
3. Computes real-time composite cyber risk scores on a calibrated 0–100 scale.
4. Translates technical CVSS scores into model-based financial exposure estimates (■8L–■15L prototype).
5. Simulates dynamic what-if wargame scenarios with instant risk and liability delta updates.
6. Optimizes security investment portfolios using genuine 0/1 Knapsack Dynamic Programming ($\sum \text{cost}_i \leq W$).
7. Maintains a tamper-evident SHA-256 Merkle audit chain with live tamper detection verification.
8. Provides conversational risk intelligence and co-pilot guidance via Groq LPU LLM inference.

NEXT EVOLUTION: Formal FAIR Monte Carlo distributions, NIST CSF 2.0 / SP 800-30 mapping, CIS Controls v8 alignment, and Bayesian predictive AI.

HOW TO VERIFY LIVE IN 60 SECONDS:

- **Launch Prototype:** Run `.\run_aura_mvp.bat` or visit <http://localhost:5173>
- **Test 0/1 Knapsack DP:** Adjust budget slider (■3L–■15L) to observe real-time $O(N \cdot W)$ dynamic allocation.
- **Run Unit Tests:** Execute `python -m unittest test_knapsack.py` in Backend/ (10 tests pass in 0.006s).
- **Inspect Blockchain Ledger:** Toggle 'Simulate Telemetry Tampering' to view cryptographic hash breaks.
- **GitHub Repository:** <https://github.com/aryanbaghel756-spec/AURA-SENTINEL>